

Q1: 24 Feb (Shift 1) - Single Correct

Let p and q be two positive number such that $p + q = 2$ and $p^4 + q^4 = 272$. Then p and q are roots of the equation :

(1) $x^2 - 2x + 2 = 0$

(2) $x^2 - 2x + 8 = 0$

(3) $x^2 - 2x + 136 = 0$

(4) $x^2 - 2x + 16 = 0$

Q2: 24 Feb (Shift 2) - Single Correct

Let a, b, c be in arithmetic progression. Let the centroid of the triangle with vertices (a, c) , $(2, b)$ and (a, b) be $\left(\frac{10}{3}, \frac{7}{3}\right)$. If α, β are the roots of the equation $ax^2 + bx + 1 = 0$, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is:

(1) $\frac{71}{256}$

(2) $-\frac{69}{256}$

(3) $\frac{69}{256}$

(4) $-\frac{71}{256}$

Q3: 24 Feb (Shift 2) - Numerical

The number of the real roots of the equation $(x + 1)^2 + |x - 5| = \frac{27}{4}$ is

Q4: 25 Feb (Shift 1) - Single Correct

The coefficients a, b and c of the quadratic equation, $ax^2 + bx + c = 0$ are obtained by throwing a dice three times. The probability that this equation has equal roots is :

(1) $\frac{1}{54}$

(2) $\frac{1}{72}$

(3) $\frac{1}{36}$

(4) $\frac{5}{216}$

Q5: 25 Feb (Shift 1) - Single Correct

The integer ' k ', for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in R is :

- (1) 3
- (2) 2
- (3) 4
- (4) 0

Q6: 25 Feb (Shift 2) - Single Correct

If $\alpha, \beta \in R$ are such that $1 - 2i$ (here $i^2 = -1$) is a root of $z^2 + \alpha z + \beta = 0$, then $(\alpha - \beta)$ is equal to:

- (1) 7
- (2) -3
- (3) 3
- (4) -7

Q7: 25 Feb (Shift 2) - Single Correct

Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is:

- (1) 4
- (2) 1
- (3) 2
- (4) 3

Q8: 26 Feb (Shift 1) - Numerical

The number of solutions of the equation $\log_4(x - 1) = \log_2(x - 3)$ is

Q9: 26 Feb (Shift 2) - Numerical

Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $P_n = (\alpha)^n + (\beta)^n$, $P_{n-1} = 11$ and $P_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of P_n^2 is

Answer Key

Q1 (4)

Q2 (4)

Q3 (2)

Q4 (4)

Q5 (1)

Q6 (4)

Q7 (3)

Q8 (1)

Q9 (324)

Q1 (4)

$$(p^2 + q^2)^2 - 2p^2q^2 = 272$$

$$((p + q)^2 - 2pq)^2 - 2p^2q^2 = 272$$

$$16 + 16pq + 2p^2q^2 = 272$$

$$(pq)^2 - 8pq - 128 = 0$$

$$pq = \frac{8 \pm 24}{2} = 16, -8$$

$$pq = 16$$

Now

$$x^2 - (p + q)x + pq = 0$$

$$x^2 - 2x + 16 = 0$$

Q2 (4)

$$2b = a + c$$

$$\frac{2a+2}{3} = \frac{10}{3} \text{ and } \frac{2b+c}{3} = \frac{7}{3}$$

$$a = 4, \left. \begin{array}{l} 2b + c = 7 \\ 2b - c = 4 \end{array} \right\}, \text{ solving}$$

$$b = \frac{11}{4}$$

$$c = \frac{3}{2}$$

$$\therefore \text{Quadratic Equation is } 4x^2 + \frac{11}{4}x + 1 = 0$$

$$\therefore \text{The value of } (\alpha + \beta)^2 - 3\alpha\beta = \frac{121}{256} - \frac{3}{4} = -\frac{71}{256}$$

Q3 (2)

$$x \geq 5$$

$$(x + 1)^2 + (x - 5) = \frac{27}{4}$$

$$\Rightarrow x^2 + 3x - 4 = \frac{27}{4}$$

$$\Rightarrow x^2 + 3x - \frac{43}{4} = 0$$

$$\Rightarrow 4x^2 + 12x - 43 = 0$$

$$x = \frac{-12 \pm \sqrt{144 + 688}}{8}$$

$$x = \frac{-12 \pm \sqrt{832}}{8} = \frac{-12 \pm 28.8}{8}$$

Hints and Solutions

MathonGo

$$= \frac{-3 \pm 7.2}{2}$$

$$= \frac{-3+7.2}{2}, \frac{-3-7.2}{2} \text{ (Therefore no solution)}$$

For $x \leq 5$

$$(x+1)^2 - (x-5) = \frac{27}{4}$$

$$x^2 + x + 6 - \frac{27}{4} = 0$$

$$4x^2 + 4x - 3 = 0$$

$$x = \frac{-4 \pm \sqrt{16+48}}{8}$$

$$x = \frac{-4 \pm 8}{8} \Rightarrow x = -\frac{12}{8}, \frac{4}{8}$$

$\therefore$ 2 Real Root's

Q4 (4)

$$ax^2 + bx + c = 0$$

$$a, b, c \in \{1, 2, 3, 4, 5, 6\}$$

$$n(s) = 6 \times 6 \times 6 = 216$$

$$D = 0 \Rightarrow b^2 = 4ac$$

$$ac = \frac{b^2}{4} \text{ If } b = 2, ac = 1 \Rightarrow a = 1, c = 1$$

$$\text{If } b = 4, ac = 4 \Rightarrow a = 1, c = 4$$

$$a = 4, c = 1$$

$$\text{If } b = 6, ac = 9 \Rightarrow 2, c = 2$$

$$\therefore \text{ probability} = \frac{5}{216}$$

Q5 (1)

$$D < 0$$

$$(2(3k-1))^2 - 4(8k^2-7) < 0$$

$$4(9k^2-6k+1) - 4(8k^2-7) < 0$$

$$k^2 - 6k + 8 < 0$$

$$(k-4)(k-2) < 0$$

Hints and Solutions

MathonGo

$$2 < k < 4$$

$$\text{then } k = 3$$

Q6 (4)

$$(1 - 2i)^2 + \alpha(1 - 2i) + \beta = 0$$

$$1 - 4 - 4i + \alpha - 2i\alpha + \beta = 0$$

$$(\alpha + \beta - 3) - i(4 + 2\alpha) = 0$$

$$\alpha + \beta - 3 = 0 \quad \&4 + 2\alpha = 0$$

$$\alpha = -2 \quad \beta = 5$$

$$\alpha - \beta = -7$$

Q7 (3)

$$x^2 - 6x - 2 = 0 \begin{cases} \alpha \\ \beta \end{cases} \quad \begin{aligned} \alpha + \beta &= 6 \\ \alpha\beta &= -2 \end{aligned}$$

$$\text{and } \alpha^2 - 6\alpha - 2 = 0 \Rightarrow \alpha^2 - 2 = 6\alpha$$

$$\text{Now } \beta^2 - 6\beta - 2 = 0 \Rightarrow \beta^2 - 2 = 6\beta$$

$$\begin{aligned} \frac{a_{10} - 2a_8}{3a_9} &= \frac{(\alpha^{10} - \beta^{10}) - 2(\alpha^8 - \beta^8)}{3(\alpha^9 - \beta^9)} \\ &= \frac{(\alpha^{10} - 2\alpha^8) - (\beta^{10} - 2\beta^8)}{3(\alpha^9 - \beta^9)} \\ &= \frac{\alpha^8(\alpha^2 - 2) - \beta^8(\beta^2 - 2)}{3(\alpha^9 - \beta^9)} \end{aligned}$$

$$= \frac{\alpha^8(6\alpha) - \beta^8(6\beta)}{3(\alpha^9 - \beta^9)} = \frac{6(\alpha^9 - \beta^9)}{3(\alpha^9 - \beta^9)} = \frac{6}{3} = 2$$

Q8 (1)

$$\frac{1}{2} \log_2(x - 1) = \log_2(x - 3)$$

$$x - 1 = (x - 3)^2$$

$$x^2 - 6x + 9 = x - 1$$

$$x^2 - 7x + 10 = 0$$

Hints and Solutions

MathonGo

$$x = 2, 5$$

$X = 2$ Not possible as $\log_2(x - 3)$ is not defined

$\Rightarrow$ No. of solution = 1

Q9 (324)

Given, $\alpha + \beta = 1, \alpha\beta = -1$

$\therefore$ Quadratic equation with roots α, β is $x^2 - x - 1 = 0 \Rightarrow \alpha^2 = \alpha + 1$

Multiplying both sides by α^{n-1} $\alpha^{n+1} = \alpha^n + \alpha^{n-1} \dots(1)$

Similarly, $\beta^{n+1} = \beta^n + \beta^{n-1} \dots(2)$

Adding (1)&(2) $\alpha^{n+1} + \beta^{n+1} = (\alpha^n + \beta^n) + (\alpha^{n-1} + \beta^{n-1})$

$$\Rightarrow P_{n+1} = P_n + P_{n-1}$$

$$\Rightarrow 29 = P_n + 11 \text{ (Given , } P_{n+1} = 29, P_{n-1} = 11 \text{)}$$

$$\Rightarrow P_n = 18$$

$$\therefore P_n^2 = 18^2 = 324$$